INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

Machine Learning Assignments

Deadline: 31st October, 2026

1. Use a standard dataset and compare the following classifiers. **[10 Marks]** a. K-Nearest Neighbour Classifier with cross-validation
   b. Decision Tree
   c. Support Vector Machine with RBF Kernel
   d. Deep Convolutional Neural Network **(Bonus: 5 marks)**

   Implementation Details:
   i) Use Train-test split as 80% Train and 20% test.
   ii) Generate the confusion matrix.
   iii) Compare accuracy, precision, recall, F1-score and ROC-AUC.
   iv) Discuss the effect of overfitting and model complexity.
   v) State which classifier is best among the given classifiers based on ROC curve analysis.

2. Use a standard dataset and compare the following regression models **[10 Marks]** a. Linear regression with gradient descent optimization
   b. Polynomial Regression
   c. Ridge and Lasso Regressions

   Implementation Details:
   i) Calculate MSE, RMSE and R^2 for each regression model and compare them. ii) Show overfitting and underfitting cases.
   iii) Use cross-validation to select the regularization parameter.

3. a) Use a standard dataset and implement K-Means Clustering from scratch. **[5 Marks]**

   Implementation Details:
   i) Experiment with different numbers of clusters: K=2,3,4,5,6.
   ii) For each value of K, calculate the Within-Cluster Sum of Squares (WCSS).
   iii) Plot the Elbow curve.

   b) Implement PCA using the covariance matrix and eigenvalue decomposition. **[5 Marks]**

   Implementation Details:
   i) Calculate the eigenvalues and eigenvectors.
   ii) Determine the explained variance ratio.
   iii) Plot cumulative explained variance.
   iv) Reduce the original feature space to:
   - 2 principal components
   - 3 principal components

   v) Visualize the data using the first two principal components.

Submission: Write a report on the above assignments and upload a pdf file. Submit the codes in zip.